

# Selective Noise Cancellation using Machine Learning

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**Abstract**—Research on Adaptive Noise Cancellation has concentrated on cancelling all audio signals other than the one the user desires to hear. This paper proposes a method for using machine learning in signal processing to train the model in a manner that can cancel the noise signals. Active noise cancellation has existed for quite some time, but it completely canceled out all the frequencies. But with adaptive noise cancellation, the user should be free to choose the frequencies they want to hear. The goal of this system is to improve and take care of hearing health in noisy cities, such as for example, noise from traffic, engines from factories, reckless driving from drivers, and noise from street vendors, all while making sure the important sounds can be still audible by the listener. Previous researches have focused only on training audios based on available datasets, whereas an entire dataset of noise and its features was constructed in order to train the model. Further, a comparison is analysed for the results achieved through the selected algorithms which helps state that KNN provides which a higher accuracy of 86% over Logistic Regression.

**Keywords**— *K-Nearest Neighbor, Logistic Regression, amplitude envelope, pitch, noise, filtering, adaptive, machine learning.*

## I. INTRODUCTION

Noise is defined in the domain of audio engineering as any undesirable or random signal present in an audio transmission. The presence of noise in an audio transmission may appear as hiss, hum, crackle, and distortion, all of which have detrimental impacts on the clarity and quality of the intended audio material. Hiss is often created by electronic components, while hum results from electrical interference. Physical defects on the recording medium, such as scratching or dust, may cause crackling and popping noises. Audio signal clipping or overdriving may cause distortion.

To date, research has focused on reducing noise from every audio input, which has eliminated every audio signal, whether noise or meaningful. For instance, active noise cancellation (ANC) Headphones isolate music from other sounds. A background emergency siren is unlikely to be heard. This idea addresses this issue. Starting with datasets created by extracting features from an audio signal and marking them '1' or '0' based on whether the noise/audio signal was permitted to pass, and used the K Nearest Neighbor (KNN) and Logistic Regression (LR) Approaches to analyze them.

Audio quality gets affected by noise. Noise barriers, filters, and software help audio engineers remove noise from an audio stream, producing a clear, high-quality product. Barking dogs, car horns, and building noise might annoy. Public safety requires ambulance sirens and smoke alerts. Numerous studies have linked noise pollution to hearing loss, sleep issues, cardiovascular illness, and cognitive deficits. Noise pollution increases stress, decreases output, and lowers living quality. Thus, this study aims to reduce urban noise pollution, notably in homes, hospitals, and schools. This research effort may improve urban dwellers' well-being and quality of life by increasing the understanding of noise pollution and finding effective solutions.

Further, it is discussed the literature that was referred to, in Section II. The limitations shortlisted from the literature surveys are then listed out and mentioned in Section III in detail. Section IV provides a brief explanation of the proposed model. There were various software's and libraries made use of in order to construct this model which are listed and discussed in Section V. The Audio Features are a key component of the Proposed Model, a detailed description of all the features in use is provided in Section VI of the paper. Section VII discusses the Methodology followed by the Results and Analysis of the same in Section VIII. Further, the Future Scope and Conclusion are derived in Section IX and Section X, respectively.

## II. LITERATURE SURVEY

The research was laid out on numerous papers, derived from various repositories such as ResearchGate, IEEE, Springer, etc., that were studied as a part of the literature survey which are discussed below:

This research focused on the study of Widrow-Hoff algorithm and how it helps in adaptive noise cancellation of acoustic noise. In the case of ANC, there exists no signal to enhance and all the detected input must be cancelled. [1] Another way to link the primary input with the corrupted signal and the reference input with the noise. Noise in the primary input can be removed without causing signal distortion after the reference input becomes signal-free and certain additional conditions are met. [2] Another study focuses on applying the Fourier Transform to the input signal and achieves results comparable to those of a traditional adaptive transversal filter while claiming to need significantly less computing. [3] One of the most promising

reference papers used the Least Mean Square algorithm which is expected to provide with the desired results but has a trail of limitations along with. An automatic methodology is adopted for time varying  $q$  parameters get better efficiency and minimize the steady state error. [4] The next paper works in depth of the previous method, and is named, Kwong Least Mean Square algorithm. It proved to have fast convergence and better noise suppression ability than traditional algorithms. [5] This paper applied the traditional adaptive noise cancellation algorithm to suppress wide band noise. For a wide band noise in 0 kHz frequency range, 15- 25 dB noise suppression can be obtained using 6 subsystems on condition that the input SNR is 0 dB and the length of the adaptive filters can be increased for hardware realization. [6] Another paper focused on LMS and NLMS algorithms in the frequency domain to achieve results. As a result, a significant reduction in mathematical computation was observed. The SNR of output signal rises about 8-9 times in frequency domain than the time domain. [7] This paper focuses on some features extraction and noise reduction techniques such as Kalman Filtering and Spectral Subtraction and compares them. The study of features came in handy to derive the features that were used in further research. [8] The next paper concentrates on comparison of Recursive Least Square Algorithms, NLMS and LMS. Also, an extensive research material on Mean Square Error, Convergence Rate, Computational Complexity of all the mentioned algorithms was found. [9]

### III. GAPS IN LITERATURE SURVEY

The above summarized literature survey provided with the basic information and algorithm study that was required in order to begin the research on the features and the methods one can implement. One of the main motivations behind working on this approach was that ANC focuses on removing all other audio signal except the one a user wishes to listen to. In such cases, the sound of an ambulance siren or even an emergency alarm too will be cancelled which can prove to be hazardous in certain situations. Hence, the model works on the aim to cancel the noise signal and alongside allow signals which are necessary to be audible to the human ear. Even then, there were a few limitations found in literature survey. The limitations are as listed below:

#### A. Lack of Dataset

An obvious approach adopted in most of the literature visited, was that none of them had an established dataset that focused on the features of an audio signal, making it a unique edge that one could entirely focus on while working on the model.

#### B. Slow convergence time

The convergence time of the LMS algorithm can be slow when the SNR is low. This might result in a delay in the model before the algorithm is able to affectively cancel the noise.

#### C. Dependency on Assumptions

An assumption is made in case of these algorithms that the noise and input signal are uncorrelated. This is not a real-

world scenario. This might lead to failure of algorithm in some cases.

#### D. Sensitivity of Input Signal

All the mentioned algorithms are very sensitive to the input signals provided and hence a change in input would not be efficiently traced by the algorithm leading to error in the output.

#### E. Computational Complexity

The algorithms require a lot of computational resources, especially in cases of large dataset, which might lead to limiting its practical application.

## IV. PROPOSED MODEL

The main goal here is to overcome the limitations that are have listed above. In order to do so, the initial step was to create a dataset consisting of the audio feature values for over 1000 audio files. The features which were shortlisted are discussed further in Section V. The dataset creation process was the most crucial step as the entire model depended on the same. Instead of training the model by data of audio signals which are to be passed, this approach is focused on training it by data of the noise signals which is a newer approach to this entire study. Additionally, because KNN and LR Machine Learning models have a primary focus, the problem of the models result generation delay cannot be observed. When it comes to Dependency on Assumptions, this model works on the dataset constructed and hence audios from the real-world scenario and its values are implemented.

## V. SOFTWARE REQUIREMENTS

Numerous libraries and software's were made use of in order to construct the model. The following are as listed below:

#### A. Python 3.9

Python is a high-level programming language famous for its simplicity, readability, and adaptability. It has a vast environment and supports many programming paradigms. Python 3.9, in comparison to Python 3.8, includes several new features and improvements, including increased dictionary support and customizable annotations.

#### B. KNeighbourClassifier

KNeighbourClassifier is a supervised learning approach in scikit-learn that is used to tackle classification problems. A sample is allocated to the category that appears most frequently among its  $k$  nearest neighbors.

#### C. Audiosegment

Audiosegment is a Python module used for audio segmentation and processing. It offers tools for segmenting audio files into smaller pieces based on silence and for processing audio files with various effects, such as pitch shifting and temporal stretching. Audiosegment can also be utilized for audio transcription and speech recognition tasks.

#### D. LogisticRegression

LR is a categorization problem-solving machine learning method from the scikit-learn toolkit. With

a collection of independent variables, it is used to predict a binary outcome (1 / 0, Yes / No, True / False). The likelihood of the outcome is predicted using a logistic function, and the decision boundary is a threshold probability over which the sample is classified as belonging to one class and below which it belongs to the other class.

## VI. AUDIO FEATURES

**Audio Features** Audio features are used in machine learning to encode and extract information from audio signals, which can then be used for tasks such as classification, clustering, and retrieval. Audio features are divided into three types: temporal, spectral, and higher-level features.

### A. Amplitude Envelope

Audio processing uses the amplitude envelope (as shown in Fig 1) to measure sound wave strength over time and extract parts of an audio signal for voice recognition and music analysis. Rectifiers take the signal's pure value and create a "rectified" signal with only positive values. Smoothing filters remove high-frequency noise and shifts. The amplitude envelope remains after rectification and smoothing, showing sound volume, pitch, and rhythm. It helps identify sound signal elements.

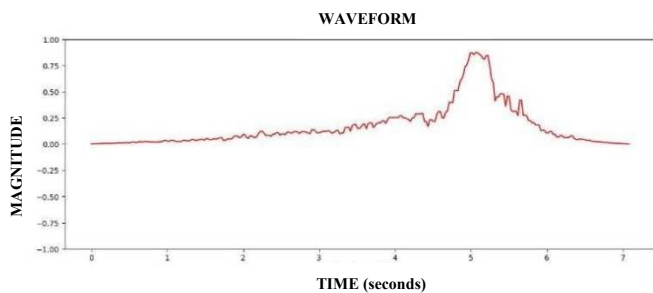


Fig. 1. Red highlight indicating the amplitude envelope of the signal

### B. Frequency

Audio data processing relies on frequency to analyze, alter, and synthesize audio signals. Fourier analysis and short-time Fourier analysis assess audio source spectral content using frequency. Cepstral or autocorrelation can also determine an audio stream's fundamental frequency, which is its observed pitch. Both frequency and pitch are essential for sound study and modification.

### C. Pitch

Pitch is an essential audio characteristic utilized in audio signal processing, especially for evaluating and modifying speech and music signals. Frequency, the basic physical attribute of sound waves that defines the number of cycles per second, is closely connected to pitch. Nevertheless, pitch is a perceptual property of sound that reflects a sound's perceived frequency, not its actual frequency.

## VII. METHODOLOGY

### A. Dataset Collection

Creating an extensive dataset was the most crucial initial step in this model research, where a wide range of audio files that are commonly considered as noise was collected. By compiling a wide variety of audios, the aim was to create a dataset that represents the types of noise that individuals may

encounter in their day-to-day lives. This dataset formed the base of the study, and was used to train and test the machine learning models which are discussed below, ensuring their ability to accurately identify and remove unwanted noise from audio recordings. The process of dataset collection is as described in Fig. 2 given below.

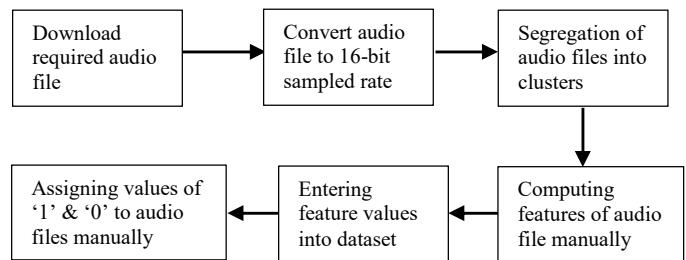


Fig. 2. Flowchart of Data Collection Process

The datasets consisted of features such as Frequency, Pitch and Amplitude Envelope. The process consisted of constructing a code to derive all the three values from an audio file manually and entering them into a sheet to further create the csv file as shown in the figure below.

TABLE 1: CSV File

| Pass/Reject | Pitch (Hz) | Amplitude Envelope |
|-------------|------------|--------------------|
| 0           | 263        | 234.21             |
| 1           | 178        | 241                |
| 1           | 403        | 126.9              |
| 0           | 70         | 252.95             |
| 1           | 78         | 212.98             |

### B. Generating Wave Plots visually

A wave plot (as shown in fig 3) is a graphical representation of the amplitude of an audio stream over time. It is generated on the Collab platform by uploading the desired audio file and importing it into the notebook using librosa.

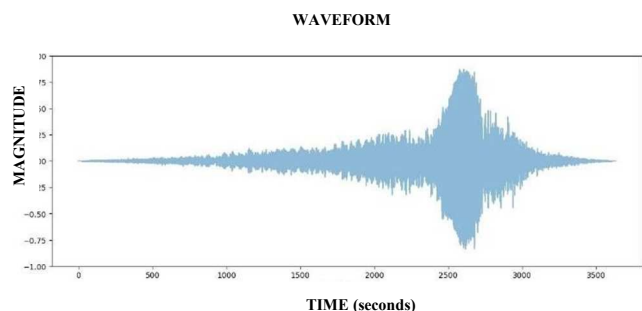


Fig. 3. Wave plot of a signal

After importing the audio data, a library such as matplotlib may be used to build the wave plot. With the "matplotlib.pyplot.plot()" method, the amplitude of an audiosignal may be graphed over time. Lastly, the wave plot may be modified to improve its readability and aesthetic appeal. Creating a wave plot in Collab involves importing the audio file, loading the audio data, making the plot, and modifying the plot. This can be done by labelling the x and y axes, modifying the color and line style, and adding a caption. The resultant wave plot can give a clear and useful picture of an audio signal's

amplitude over time.

### C. Extracting Features from Wave Plots

The features had to be manually extracted. It can be further explained: Calculating the frames, sample rate, and sample width of the audio signal is necessary before transforming it into a NumPy array. The audio signal's frequency and pitch are calculated using the FFT, with the highest frequency determined first and multiplied by the frame rate.

#### 1) Extracting Amplitude Envelope

The Python Librosa module extracts an audio signal's amplitude envelope using the amplitude to "db()" and "stft()" methods. The "librosa.load()" function and sample rate (sr) have been employed to import audio data. The audio stream's Short-time Fourier transform (STFT) is computed using the "librosa.stft()" function. The power spectral density matrix is converted into a power spectral density matrix using "librosa.amplitude" to "db()". The amplitude envelope can be found using the "np.max()" method.

#### 2) Extracting Frequency

Python uses the Fast Fourier Transform (FFT) to identify signal frequency. Python's NumPy module implements FFT efficiently. It requires creating a time-domain signal, FFT computation, deciding frequency, and plotting FFT amplitude. NumPy's "np.fft.fftfreq" method generates Hz-measured frequencies. Plotting size versus frequency shows this.

#### 3) Extracting Pitch

Python's Fast Fourier Transform (FFT) method can generate pitch by converting time-domain signals to frequency-domain ones. It requires loading music files with librosa, smoothing the audio signal, calculating the FFT output, quadrupling the FFT output, getting power spectral frequency bins, and finding the highest power spectrum score. Spectrum modelling synthesis and sine modelling can improve FFT pitch accuracy.

### D. Computing the ML Algorithms

Machine learning helps in pattern spotting, prediction, automation, personalization, and optimization. It can identify faces in photos and transcribe voice into text, predict consumer behavior and sports events, automate data entry and quality control, improve supply lines and equipment maintenance, and predict, automate, improve, and customize customer encounters. These models are effective for data analysis and pattern detection that humans cannot do directly.

#### 1) Logistic Regression

LR estimates the likelihood of a binary outcome as a function of predictor factors, which is a generalized linear model. The LR Algorithm uses Amplitude Envelope because of its uniqueness, to

segregate values and could adaptively give "1" or "0" to pass or refute the audio file.

$$\log(odds) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 \dots + \beta_p x_p \quad \dots (1)$$

where,  $\beta_0$  indicates the log-odds of the dependent variable when all independent variables are equivalent to zero.  $\beta_1, \beta_2, \dots, \beta_p$  are the coefficients of the independent variables  $x_1, x_2, \dots, x_p$ , respectively, in Equation 1. These coefficients represent the change in log-odds of the dependent variable for a one-unit rise in the corresponding independent variable, all other independent variables kept constant. For the purpose of LR, the independent variables that are used are the characteristics of the audio signal, more specifically the amplitude envelope value.

#### 2) K – Nearest Neighbor

KNN is a powerful and flexible algorithm that can be a useful tool in the data scientist's toolkit. However, value of k should be carefully considered, the choice of distance metric, and potential issues with missing data and feature scaling. It is a simple and intuitive algorithm that can very often produce highly accurate results, as this one, with little to no data preprocessing required.

Let  $D = (x_1, y_1), (x_2, y_2), (x_n, y_n)$  be the training dataset for a given data point  $x$ , where  $x_i$  denotes the  $i$ -th data point and  $y_i$  denotes its corresponding class label.

The algorithm operates as follows: initially, the value of  $k$ , which represents the number of neighbors closest in proximity to be considered, is determined. After that, the distance that separates the newly added data point  $x$  from the rest of the data points in  $D$  was calculated. Furthermore, the calculated distances are used to choose the  $k$  members who are the closest neighbors.

Using the Euclidean distance formula, the distance between two data points  $x$  and  $x_i$  can be calculated as follows:

$$d(x, x_i) = \sqrt{(x_1 - x_{i1})^2 + (x_2 - x_{i2})^2 + \dots + (x_p - x_{ip})^2} \dots (2)$$

where  $p$  denotes the number of features (independent variables) in the dataset, as given in Equation 2. In this case, two audio signal features are considered, amplitude envelope and pitch.

However, KNN has a few limitations such as the computational cost involved in finding the  $k$  nearest neighbors for each new data point, especially when dealing with large datasets. In addition, the selection of an appropriate distance metric holds very crucial for the algorithm's performance.

### E. Generating Anti-Noise Wave

In audio processing, an inverse waveform which operates out of phase with the original waveform has been generated using the phase inversion technique. A shift in the polarity of a signal by reversing its phase by 180 degrees was noticed. This is typically done to create a "negative" version of a

signal in order to eliminate phase cancellation when combining two signals.

The anti-noise wave operates on the principle of destructive interference. When two sound waves with opposing phases collide, a phase shift occurs that leads in the cancellation of sound. By generating an anti-noise wave with the same amplitude, frequency, and phase as the original sound wave, the anti-noise wave effectively cancels out the original sound wave, creating a quieter environment.

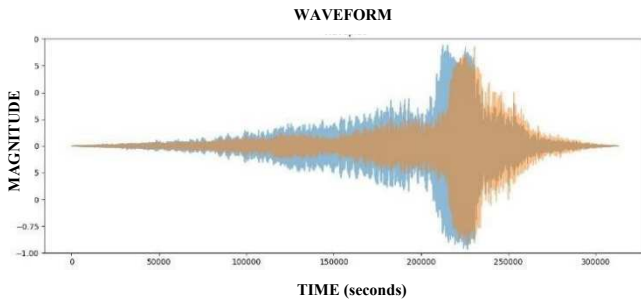


Fig. 4. Plot of Anti-Noise Waveform

Consider a digital audio signal, as given in Equation 3,

$$s[n] = (A \times \sin(2\pi f_n \Delta t + \varphi)) \dots (3)$$

where:

- $s[n]$  : value of the audio signal at the  $n$ -th sample.
- $\Delta t$ : sampling interval, which represents the time between consecutive samples.
- $F_n$ : frequency of the audio signal in hertz (Hz),

which is equal to  $f/\Delta t$ .

Then the corresponding anti-phase signal will be as follows in Equation 4,

$$s[n] = -(A \times \sin(2\pi f_n \Delta t + \varphi)) \dots (4)$$

## VIII. RESULTS AND ANALYSIS

The following are the results of both the applied algorithms. For the same various libraries of Python, namely, sklearn, pydub and librosa was used.

Sklearn provided a wide range of tools for classification, regression, clustering tasks. Similarly, Librosa provided tools for feature extraction, beat tracking and spectrogram visualization. It was majorly used for analyzing and processing audio signals. Pydub was useful in manipulating the audio files and provided with an interface to read variety of audio formats which were required to derive the results below.

TABLE 2: RESULT AND ANALYSIS OF LR

|        | Precision | Recall | F1-Score | Support |
|--------|-----------|--------|----------|---------|
| Reject | 68        | 79     | 73       | 19      |
| Pass   | 64        | 50     | 56       | 14      |

TABLE 3: RESULT AND ANALYSIS OF KNN

|        | Precision | Recall | F1-Score | Support |
|--------|-----------|--------|----------|---------|
| Reject | 88        | 95     | 91       | 22      |
| Pass   | 89        | 73     | 80       | 11      |

The results indicate that the precision for class 0 [the frequencies desired to eliminate] is 0.75, indicating that 75% of the predicted class 0 samples were correctly classified. The recall is 1.00, which indicates that 100 percent of the genuine class 0 samples were correctly classified. The F1 score of 0.86 represents the harmonic mean of precision and recall. This result shows a balanced indicator of both accuracy and recall.

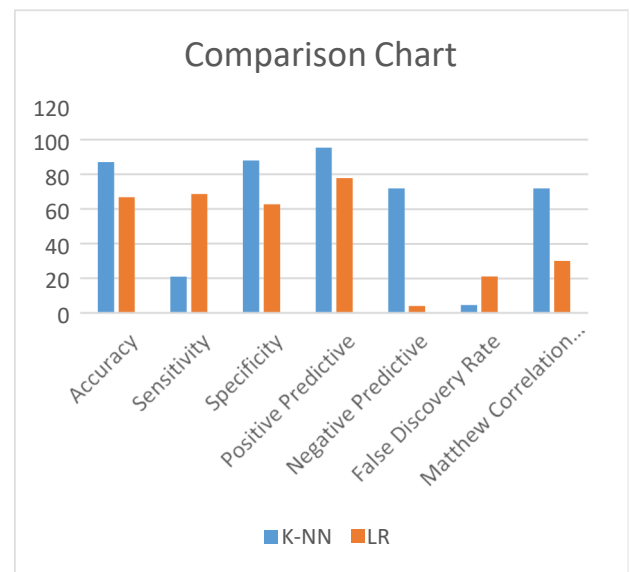


Fig 5: Comparison Chart of LR and KNN

## IX. ADVANTAGES, DISADVANTAGES, APPLICATION

### A. Advantages

ML-based noise cancellation technology's self-learning algorithm improves selectivity without repeated frequency selection. This technique improves audio without external noise. Consumers can also save money with the model's low price. This method can compute noise cancellation in real time. This noise cancelling model has many benefits, making it a common choice.

### B. Disadvantages

Implementing this system requires some important considerations. This method addresses most literary survey issues. It has specific requirements. First, the model needs a reliable power supply to work. If hardware or software issues arise, it may need expert maintenance. 16-bit audio and WAV file support are also necessary. These requirements make sure the model can handle audio inputs well. To use the adaptive

noise cancelling model effectively, you must grasp its unique needs.

### C. Implications of Model

This model once successfully complete can be used in various places, as follows:

#### i. Luxury Cars:

Here, specifically luxury Cars as they are already equipped with high-definition speakers. Only requirement will be to add High quality Microphones on the exterior to capture various frequencies. When in traffic or any other place this system can be used to filter out loud and harsh sound and give the passenger inside a very relaxed journey.

#### ii. House Balcony:

In residential flats or rooms with a lot of noise pollution, such a system allows natural frequencies like birds, emergency sirens, and rain to flow through.

#### iii. Aviation:

ANC can be utilized to reduce the impact of interior noise on aircraft passengers. This can enhance the comfort of passengers and reduce the risk of hearing loss among flight crews. It can also help pilots hear radio communications more clearly.

## X. CONCLUSION

This paper proposes, a novel approach to adaptive and selective signal processing for optimizing noise cancellation using machine learning. This system aims to improve hearing health in urban environments with excessive noise by allowing users to choose which frequencies to perceive. The system accomplishes this by employing a combination of active noise cancellation and adaptive filtering techniques to cancel out unwanted noise while preserving essential noises. This research demonstrates that this approach is effective in reducing the impact of noise pollution on hearing health. In addition, this system has great potential for practical applications in noisy urban environments, such as protecting factory employees and enhancing the commuter experience on public transportation. Overall, this research makes a significant contribution to the fields of adaptive signal processing and noise cancellation and can inspire further progress in these areas.

## XI. FUTURE SCOPE

ANC was first used in the 1950s and was based on patents by Lawrence Fogel, who is credited with developing the concept. The method was initially developed and applied for aviation, making it quieter for pilots of aero planes and helicopters. The ANC market has grown significantly both abroad and in India due to rising consumer expenditure and rising demand for high-quality audio. Also, it will be expected that ANC technology will play a significant role in the future and end up becoming a standard part of portable audio equipment. This model appears luxurious in the present day. One may think of using ANC for their own comfort, but as was mentioned above, in the approaching years, one can expect to

see ANC used in nearly every setting, including homes, schools, nursing homes, theatres, hospital rooms, and much more.

Another feature that can be added is Mel Frequency Cepstral Coefficient. The MFCC feature extraction technique consists of windowing the signal, performing the DFT, calculating the log of the magnitude, and then warping the frequencies on a Mel scale before performing the inverse DCT.

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